Chapter 22 : Communication

1. What is Communication?

- Agents exchange information using signals such as natural language, speech, gestures, or other messages.
- Communication allows agents to share beliefs, goals, intentions, and requests.
- It involves both producing messages and interpreting messages from others.

Key Idea: Communication is the process of creating, transmitting, and interpreting messages to achieve some purpose.

2. Why Communication Matters in AI

- Connects perception, knowledge, reasoning, and action.
- Enables coordination among multiple agents.
- Allows agents to acquire new knowledge from others (e.g., asking questions).
- Supports collaboration, negotiation, and task planning.
- Essential for human-AI interaction and for AI to operate in social environments.

Communication turns isolated agents into cooperative, knowledgeable, social agents.

3. Levels of Language Analysis

Level	What it Studies
Phonology	Sound structure of language (e.g., phonemes).
Morphology	Structure of words (e.g., roots, prefixes, suffixes).
Syntax	Grammatical structure of sentences (e.g., parse trees).
Semantics	Literal meaning of expressions (e.g., logical form).
Pragmatics	Meaning in context (e.g., speaker intentions, implicatures).
Discourse	Structure across multiple sentences (e.g., coherence, coreference).

4. Communication Pipeline

Input signal (text / speech) → **Lexical analysis**
- Tokenize input
- Identify words
- Handle unknown words

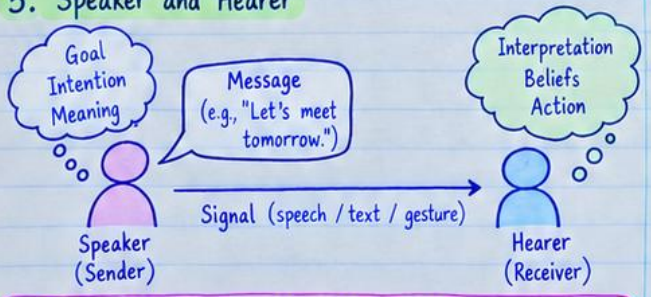
→ **Parsing (syntax analysis)**
- Build grammatical structure (parse tree)
- Check syntactic well-formedness

→ **Semantic interpretation**
- Map to meaning representation
- Build logical form or other semantic representation

→ **Discourse & Pragmatics**
- Use context and world knowledge
- Resolve references
- Infer speaker intent
- Handle implicatures

→ **Response / Action**
- Generate a reply
- Update knowledge
- Perform requested action

5. Speaker and Hearer



- Speaker selects a message to achieve a goal.
- Hearer interprets the message using linguistic knowledge, context, and shared knowledge.
- Successful communication depends on shared background knowledge and correct interpretation of intentions.

6. Meaning vs Form

Same meaning, different form

- "I am hungry."
- "I want food."
- "Could we get something to eat?"

} Different wording, same underlying meaning (request for food).

Same form, different meaning

- "Can you pass the salt?"
- "Can you open the door?"
- "Can you help me?"

} Same wording, different meaning depending on context and situation.

7. Sources of Difficulty in Communication

Source	Description	Example
Ambiguity	Multiple possible meanings (syntax or semantics).	"I saw the man with the telescope." (who has the telescope?)
Noise	Errors in signal (speech recognition, typos, etc.).	"meet" → recognized as "meat"
Incomplete information	Missing context or background knowledge.	"It is ready." (ready for what?)
Indirectness	Meaning not stated literally (e.g., implicature, politeness).	"It's getting late." (implicit suggestion to leave)
Reference resolution	Determining what expressions refer to.	"John met Bill. He was tired." (who is "He"?)

8. Real-World Applications

- Dialogue systems: conversational agents (e.g., chatbots).
- Question answering: answer questions from text or speech.
- Machine translation: translate between languages.
- Voice assistants: understand and execute spoken commands (e.g., Siri, Alexa).
- Summarization: generate concise summaries of text.
- Information extraction: find facts and relations in text.
- Multi-agent communication: coordination and negotiation (e.g., autonomous robots, distributed systems).

9. Mini Takeaway

Communication requires both structure (language rules) and context (world knowledge, situation, speaker intent).

10. Key Takeaways

1. Agents communicate to share information, goals, and requests.
2. Communication links perception, knowledge, reasoning, and action.
3. Language has multiple levels of structure and meaning.
4. Interpretation depends on context and shared knowledge.
5. Communication is challenging due to ambiguity, noise, and incomplete information.
6. Many important AI applications rely on effective communication.

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1. Grammars and Syntax

- A grammar specifies the syntactic structure of a language (set of well-formed sentences).
- A formal grammar consists of a set of symbols, a start symbol, and a set of production rules.
- Syntax determines how words can be combined to form valid sentences.

Key Idea: Grammars define the structure of sentences, independent of their meaning.

2. Context-Free Grammar (CFG)

- A CFG uses production rules of the form $A \rightarrow \alpha$, where A is a nonterminal and α is a sequence of terminals/nonterminals.
- Commonly used to model the syntax of natural language.
- Captures hierarchical structure and allows recursive constructions.

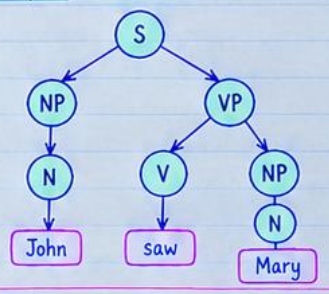
Example: A small CFG for simple sentences

S	→	NP VP
NP	→	Det N | ProperNoun
VP	→	V | V NP | V NP PP
PP	→	P NP
Det	→	the | a
N	→	man | woman | telescope
V	→	saw | liked
P	→	with | in
ProperNoun	→	John | Mary

3. Parse Trees

- A parse tree shows the hierarchical syntactic structure of a sentence according to a grammar.
- The root is the start symbol, leaves are words, and internal nodes are nonterminals.
- Each path from root to leaf shows how the sentence is generated.

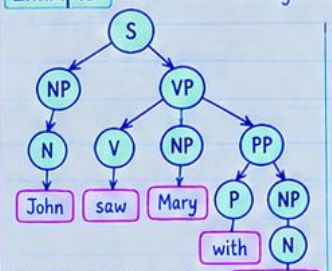
Example: Parse tree for "John saw Mary"



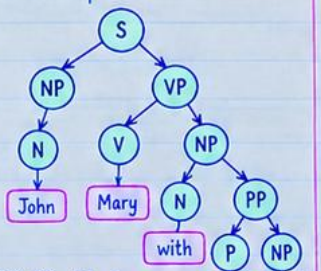
4. Syntactic Ambiguity

- A sentence may have more than one valid parse tree (syntactic ambiguity).
- Occurs when a string can be generated in multiple ways according to the grammar.
- Common in natural language due to flexible structure.

Example: "John saw Mary with a telescope"



(a) PP with the verb (saw using a telescope)



(b) PP with the noun (Mary with a telescope)

5. Semantic Interpretation

- Maps a syntactic structure (parse tree) to a meaning representation (e.g., a logical form).
- Determines what is being talked about and how entities, relations, and events are connected.
- Uses rules to compose the meaning of a sentence from the meanings of its parts.

Key Idea: Syntax defines structure; semantics defines meaning.

6. Compositionality

- The meaning of a complex expression is a function of the meanings of its parts and the way they are combined.
- Principle of compositionality: the same parts have the same meanings wherever they occur.
- Allows systematic interpretation of new sentences.

Example:

$$\text{meaning}(\text{saw}(\text{John}, \text{Mary})) = \text{see}(\text{John}, \text{Mary})$$
$$\text{meaning}(\text{John saw a dog}) = \text{see}(\text{John}, \text{dog})$$

7. Lexical Ambiguity

- A word can have multiple possible meanings (lexical ambiguity).
- Requires word-sense disambiguation using context.
- The same syntactic structure may yield different meanings depending on the word sense.

Example: "bank"

- river bank → meaning: land alongside a river
- financial bank → meaning: financial institution

8. Semantic Roles

- Semantic roles describe the function that an entity plays in an event (e.g., agent, patient, instrument, location).
- Help capture "who did what to whom, how, when, and where" independent of surface syntax.
- Useful for deeper language understanding and reasoning.

Example: "John saw Mary with a telescope"

Agent	:	John
Patient	:	Mary
Instrument	:	a telescope

9. Additional Notes

- Grammars capture the set of possible sentences, not their probabilities.
- Natural language meaning is often underspecified and context-dependent.
- Pragmatics (use, intent, context) is needed for full interpretation.
- Real-world systems combine syntax, semantics, and context for robust understanding.

10. Key Takeaways

1. Grammars define the syntactic structure of language.
2. Context-free grammars are widely used for natural language.
3. Parse trees represent hierarchical structure.
4. Sentences can be syntactically ambiguous.
5. Semantic interpretation maps structure to meaning.
6. Compositionality builds meaning from parts.
7. Words can have multiple meanings (lexical ambiguity).
8. Semantic roles capture the function of entities in events.
9. Combining syntax, semantics, and context is essential for understanding natural language.

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1. Language Understanding as Inference

- Understand the meaning of an utterance by inferring the most likely interpretation.
- Combine linguistic knowledge with world knowledge and context using probabilistic inference.
- Resolve ambiguities, missing information, and implicit meaning.

Key Idea: Language understanding is inference to get from the surface form of an utterance to its intended meaning.

Example: "I saw the man with the telescope."
Two possible meanings:
- I used a telescope to see the man.
- I saw a man who had a telescope.

2. Ambiguity Resolution

- Natural language is often ambiguous (syntactic, semantic, pragmatic).
- Use context, world knowledge, and probabilistic models to prefer the most plausible interpretation.
- Disambiguation can be viewed as choosing the interpretation with the highest probability (MAP).

Types of Ambiguity:
- Lexical: multiple word senses (e.g., "bank" = river or financial).
- Syntactic: multiple parse structures.
- Semantic: multiple meanings.
- Pragmatic: depends on context, speaker intent, and situation.

3. Natural-Language Generation (NLG)

- Generate natural language utterances from an internal representation of meaning.
- Choose the content to express and how to express it in fluent, grammatical language.
- Uses knowledge of language structure, lexicon, and discourse.

Key Idea: NLG maps a meaning representation to a natural language sentence or dialogue.

Example: Meaning representation (inform)
goal: tell(location(paris))
↓
Generated sentence:
"Paris is located in France."

4. Generation Pipeline

Communicative Goal (meaning) → **Content Determination** → **Sentence Planning** → **Surface Realization** → **Natural Language Utterance**

- Content Determination: What to say? Select relevant information. Plan discourse structure.
- Sentence Planning: Choose phrasing. Lexical selection (word choice). Generate syntactic structure.
- Surface Realization: Produce grammatical sentence. Inflect words. Handle agreement and morphology.
- Natural Language Utterance: "The meeting is at 10 am tomorrow."

5. Machine Translation (MT)

- Translate text from a source language to a target language.
- Uses statistical or neural models to handle differences in grammar, word order, and idioms.
- Must deal with ambiguity and noisy input.

Example (noisy translation):
English : "The book is on the table."
Correct MT: "Le livre est sur la table."
Noisy MT : "Le livre est dans la table."
(literally: "in the table", incorrect)

6. Information Extraction (IE)

- Automatically extract structured information from unstructured text.
- Identify named entities (people, organizations, locations) and relations between them.
- Often formulated using sequence labeling and classification models.

Example: Text: "Barack Obama was born in Hawaii."
↓
Extracted information:
Person: Barack Obama
Relation: born_in
Location: Hawaii

7. Question Answering (QA)

- Answer natural language questions using information from text or knowledge bases.
- Requires language understanding, information retrieval, and inference.
- Questions can be factoid, list, definition, comparison, or complex reasoning.

Example:
Question: "Who wrote the book The Hobbit?"
Approach: 1. Analyze question.
2. Retrieve relevant passages.
3. Extract answer span.
4. Rank candidate answers.
Answer: "J.R.R. Tolkien"

8. Dialogue Systems

- Engage in multi-turn conversation with users to achieve goals (e.g., information seeking, booking, task assistance).
- Maintain a dialogue state (user intent, slots, context).
- Use natural language understanding (NLU) and generation (NLG) components.
- Handle clarification, mixed initiative, and error recovery.

Basic Dialogue Architecture:



9. Applications of Language Communication

Application	Task	Example
Machine Translation	Translate text between languages	"Hello" → "Bonjour"
Information Extraction	Extract entities and relations	Person: Elon Musk, Org: SpaceX
Question Answering	Answer natural language questions	Q: "When was AI founded?" A: "1956"
Dialogue Systems	Conversational agents	Chatbots, virtual assistants
Text Summarization	Generate concise summaries	News article → short summary
Text Generation	Write descriptions or reports	Generate a weather report
Spoken Language Systems	Speech recognition and synthesis	Voice assistants (e.g., Siri, Alexa)

10. Key Takeaways

1. Understanding language involves inference over meaning, using context and world knowledge.
2. Ambiguity resolution chooses the most plausible interpretation.
3. Natural language generation converts meaning to fluent, grammatical text.
4. Machine translation handles cross-language differences and noise.
5. Information extraction finds structured information from text.
6. Question answering combines language understanding and information retrieval.
7. Dialogue systems maintain context and generate multi-turn interactions.
8. Language communication has many real-world applications and is a key area in modern AI.

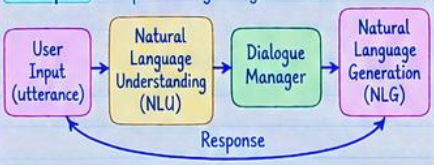
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1. Dialogue Systems

- Engage in multi-turn interaction with users.
- Combine natural language understanding, dialogue management, and natural language generation.
- Maintain a dialogue state and generate contextually appropriate responses.
- Used in chatbots, virtual assistants, and task-oriented systems.

Key Idea: Dialogue systems integrate understanding, reasoning and generation to achieve coherent and goal-directed interaction.

Example: Simple dialogue system flow



2. Dialogue Management

- Maintains dialogue state (history, user intent, slots, constraints).
- Chooses the next system action (answer, ask question, provide options, etc.).
- Handles clarification, confirmation and repair strategies.
- Balances between task success and natural, engaging conversation.

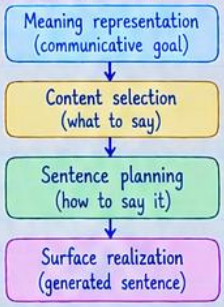
Common Dialogue Actions:

- Inform: provide information.
- Request: ask for missing information.
- Confirm: check understanding.
- Clarify: ask for clearer input.
- Suggest: provide options.
- Summarize: recap the dialogue state.
- Execute: perform requested task.
- End: close the conversation.

3. Natural-Language Generation (NLG)

- Converts a meaning representation into fluent natural language.
- Content selection: decide what to say.
- Sentence planning: organize content and determine structure.
- Surface realization: generate grammatical and natural sentences.

Example: NLG pipeline



Example:

Goal: inform about flight.

Meaning rep:

{flight = AI102, time = 10am, from = Delhi, to = Boston}

Generated sentence:

"Flight AI102 from Delhi to Boston departs at 10 am."

4. Machine Translation

- Translate text from a source language to a target language.
- Handle differences in grammar, word order, idioms and cultural expressions.
- Uses statistical or neural models.
- Faces challenges due to ambiguity, context, and multiple valid translations.

Example: Machine translation

Source (English)	→	Target (French)
"The book is on the table."	→	"Le livre est sur la table."

Challenges:

- Ambiguous words, idioms, culture-specific expressions, and long-range dependencies.

5. Information Extraction

- Automatically extract structured information from unstructured text.
- Identify named entities (people, organizations, locations, dates), relations and events.
- Uses sequence labeling, classification and deep learning models.
- Useful for knowledge base construction, search and question answering.

Example: Extracting entities and relations

"Barack Obama was born in Hawaii and served as the 44th President of the United States."

Type	Extracted Information
Person	Barack Obama
Location	Hawaii, United States
Relation	born_in(Barack Obama, Hawaii)
Relation	president_of(Barack Obama, United States)

6. Question Answering (QA)

- Answer questions using information from text or knowledge bases.
- Requires language understanding, information retrieval, and inference.
- Can be factoid (short answer) or complex (reasoning).
- Often uses reading comprehension and retrieval-augmented models.

Example: Question answering

Context: "Barack Obama was born in Hawaii and served as the 44th President of the United States."

Question: "Where was Barack Obama born?"

Answer: "Hawaii"

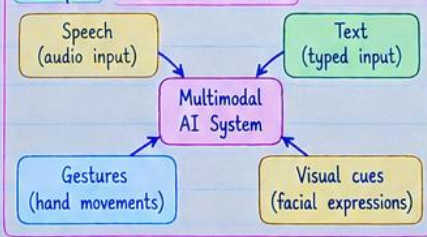
Types of Questions:

- Factoid: short, specific answer (e.g., Who?).
- Definition: meaning of a term.
- List: multiple answers (e.g., name the cities).
- Complex: requires reasoning or multiple facts.

7. Multimodal Communication

- Use multiple modalities: speech, text, gesture, facial expressions and visual input.
- Combine information from different sources for better understanding.
- Useful in human-robot interaction, virtual assistants, and accessible systems.

Example: Multimodal interaction



8. Common Challenges in Practice

- Ambiguity: multiple possible meanings.
- Ellipsis: missing words ("Book the 5 pm one.").
- Reference resolution: resolving pronouns and definite descriptions.
- Indirect speech: understanding requests and implicit meaning.
- Noisy input: speech recognition errors, typos, informal language.
- Context dependence: meaning depends on dialogue history and world knowledge.
- Domain variability: different vocabulary and styles across domains.

Example: Ambiguity and context

User: "I saw the man with the telescope."
Meaning 1: I used a telescope to see the man.
Meaning 2: I saw a man who had a telescope.
Context and world knowledge help disambiguate.

9. Practical AI Applications

Application	Description / Example
Chatbots	Task-oriented dialogue systems (e.g., airline booking).
Virtual assistants	Multi-domain systems (e.g., Siri, Alexa, Google Assistant).
Machine translation	Translate text across languages (e.g., Google Translate).
Information extraction	Extract entities and facts from news, documents and web pages.
Question answering	Answer factual and complex questions (e.g., search engines).
Text summarization	Generate concise summaries of documents or articles.
Customer support	Automated help desks and FAQ systems.

10. Key Takeaways

1. Dialogue systems integrate understanding, state tracking and generation for effective interaction.
2. NLG converts meaning into fluent language using content selection, planning and realization.
3. Machine translation, information extraction and question answering enable real-world language applications.
4. Multimodal communication combines speech, text and visual cues.
5. Practical systems face challenges such as ambiguity, reference resolution, noisy input and context dependence.
6. Communication technologies power chatbots, assistants, translation tools, search and customer support.
7. Effective communication is essential for human-AI interaction.
8. Combining multiple techniques leads to more robust and usable systems.

Chapter 22 : Communication

1. Chapter Summary

- Communication is the process of creating, transmitting, and interpreting messages between agents.
- It involves language, meaning, context, and the goals of the speaker and hearer.
- Language is studied at multiple levels: syntax, semantics, pragmatics, and discourse.
- Effective communication in AI requires integrating knowledge, reasoning, and context, and is essential for interactive and cooperative agents.

Key Idea: Communication enables agents to share information and achieve coordinated action.

2. Comparison of Levels

Level	What it Studies	Key Questions	Example
Syntax	Grammatical structure of sentences.	Is the sentence well-formed?	"The robot moves to the room." (syntactically correct)
Semantics	Literal meaning of expressions.	What does the sentence mean?	"The robot moves to the room." (refers to an action and a location)
Pragmatics	Meaning in context.	What is the speaker's intent?	"Can you close the door?" (request, not a question about ability)
Discourse	Structure across multiple sentences.	How do sentences fit together?	"I'm hungry, Can you open the fridge?" (coherent dialogue)

3. Key Terms

- **Communication** : exchange of information between agents.
- **Language** : formal system of symbols and rules.
- **Syntax** : structure of expressions.
- **Semantics** : literal meaning of expressions.
- **Pragmatics** : meaning in context.
- **Discourse** : structure across multiple sentences.
- **Speech act** : type of communicative action (e.g., inform, request).
- **Reference** : link between expressions and real-world entities.
- **Intention** : goal of the speaker.
- **Context** : situational and background knowledge.
- **Interpretation** : deriving meaning from an utterance.

4. From Utterance to Response (Recap)

Utterance (input signal) → **Syntactic analysis** → **Semantic interpretation** → **Pragmatic interpretation** → **Response / Action**

- Utterance (input signal)
  - Speech or text
  - Sequence of words
  - Produced by speaker
- Syntactic analysis
  - Parse the structure
  - Check well-formedness
- Semantic interpretation
  - Map to meaning representation
  - Identify referents
  - Combine with world knowledge
- Pragmatic interpretation
  - Use context
  - Infer speaker intent
  - Resolve ambiguity
  - Consider discourse history
- Response / Action
  - Generate reply
  - Update knowledge
  - Perform action
  - Continue dialogue

5. Syntax vs. Semantics vs. Pragmatics vs. Discourse

Syntax
- Concerned with well-formed sentences.
- Independent of meaning.
- Can be evaluated formally.
- Handled using grammars (e.g., CFGs).

Semantics
- Assigns meaning to expressions.
- Links language to the world.
- Often represented with logical forms.
- Enables truth conditions and inference.

Pragmatics
- Meaning in context.
- Accounts for speaker intent.
- Uses background knowledge and situational context.
- Explains indirect and implied meanings (e.g., requests).

Discourse
- Structure over multiple sentences.
- Maintains coherence.
- Tracks context and discourse state.
- Resolves references (e.g., pronouns).
- Important for dialogue and longer texts.

6. Common Mistakes / Pitfalls

- Confusing syntax with semantics (well-formed ≠ meaningful).
- Ignoring pragmatic interpretation (e.g., treating "Can you ...?" as a real question).
- Assuming literal meaning is always intended.
- Not using context or background knowledge.
- Failing to resolve references (e.g., pronouns).
- Ignoring discourse structure in multi-sentence input.
- Overlooking speaker intentions and social context.
- Assuming communication is only about language (ignores gestures, actions, and multimodal signals).

7. Likely Exam Questions

- Explain the different levels of language analysis (syntax, semantics, pragmatics, discourse) with examples.
- Distinguish between literal and intended meaning with suitable examples.
- Describe how context affects interpretation.
- What are speech acts? Give examples.
- Explain reference resolution and its challenges.
- How do discourse models help in communication?
- Discuss the role of world knowledge in pragmatic interpretation.
- Compare symbolic and statistical approaches to language understanding (as covered in Ch. 22).

8. Study Tips

- Make a table of examples for syntax, semantics, pragmatics, and discourse.
- Practice analyzing real sentences at different levels.
- Think about multiple possible interpretations for ambiguous utterances.
- Work through dialogue examples and track context and intentions.
- Relate concepts to real-world applications (e.g., chatbots, voice assistants).
- Focus on how different levels work together in a full communication pipeline.

9. Final Takeaways

- Communication is central to intelligent agents and multi-agent interaction.
- Language has multiple levels: syntax, semantics, pragmatics, and discourse.
- Meaning depends not only on words but also on context, speaker intent, and world knowledge.
- Effective communication requires integrating all levels.
- Understanding communication is essential for building systems that can interact naturally and cooperatively with humans and other agents.

"Communication turns isolated agents into cooperative, knowledgeable, social agents." — AIMA (Ch.22)